Rahul Krishnan

Independent AI Safety Researcher
Universitätsring 8E, 54296 Trier, Germany

+49 15510269942, rahulkrishnan1105@gmail.com, GitHub, LinkedIn, Website

Profile

Independent AI safety researcher and M.Sc. Data Science candidate at Universität Trier investigating reward misspecification, chain-of-thought faithfulness, and activation-based safety monitoring. Studied specification gaming under GRPO, pre-reasoning probes for misleading-hint following, and probe reliability under KV-cache compression. First author of two research papers on LLM inference, with four years of production software engineering experience at SAP Labs India.

Projects

- **FEAR: Reward Misspecification and Specification Gaming**

Mar - Oct 2026

- Built a GRPO training and evaluation pipeline using TRL, vLLM and LoRA; trained Qwen2.5 1.5B across eight reward configurations, with selected 3B experiments, to investigate whether stronger penalties improve calibration and selective abstention across six benchmarks.
- Found specification gaming under asymmetric rewards: increasing wrong-answer penalties raised MATH skip rates from 14% to 61% without improving skip selectivity, while accuracy on answered questions fell from 61.8% to 51.3%. At most 0.5% of skips expressed uncertainty; models instead avoided grading through code, unboxed answers, or truncation.
- Showed that GRPO's within-group reward normalization reduces asymmetric reward schemes to a single effective abstention parameter; verified this equivalence on 12,638 real rollout groups.
- Trained linear probes on model hidden states to predict answer correctness (AUROC 0.78 on MATH, 0.72 on GSM8K). Probe-guided abstention improved accuracy on answered GSM8K questions by 2.9-11.0 percentage points over penalty-trained models at matched answer rates. Packaged the abstention benchmark as a UK AISI Inspect evaluation.

- **Safety Probes under KV-Cache Compression**

Sept 2026

- Investigated whether KV-cache compression compromises LLM refusal behaviour and activation-based safety monitoring. Evaluated frozen linear safety probes under JoLT low-rank compression and SnapKV token eviction across four Qwen3 models (0.6B–8B), five compression ratios (2x-8x), and 500 held-out harmful and harmless prompts.
- Found that safety-monitor reliability depends strongly on the compression method: SnapKV preserved harmful-prompt detection across all tested ratios, while JoLT progressively degraded probe performance. At 5x, the probe still detected 96.9% of harmful prompts despite degraded model behaviour.
- Investigated compression-induced failures using error-matched Gaussian noise, token-location counterfactuals, and probe-retraining controls. Found that low-rank compression caused substantially more generation failures than random perturbations of equal reconstruction error (82.0% vs. 0.6% degenerate outputs on Qwen3-8B at 8× compression). Audited LLM safety-judge labels to identify unreliable harmful-compliance assessments.

- **Chain-of-Thought Faithfulness and Hint Probes**

Apr 2025 - Jun 2025, Oct 2026

- Co-developed a counterfactual faithfulness evaluation across Gemini 2.5 Flash Lite and LLaMA 3.2 3B on 650 BIG-Bench Hard instances. Reflective CoT reduced bias susceptibility by approximately 7 percentage points but did not eliminate unfaithfulness: revisions sometimes reinforced biased answers with plausible but misleading explanations.
- Trained residual-stream linear probes on Qwen3 0.6B–8B to predict whether models would follow misleading hints before generating a response. On held-out reasoning tasks, probes at 4B and 8B improved prediction by 0.07 AUROC beyond baselines using prompt features or direct answers.
- Tested whether predictive representations causally influence hint-following using counterfactual prompts and pre-registered activation steering with random-direction controls. Found no significant effect from probe-direction steering at the tested strength; packaged the benchmark as a UK AISI Inspect evaluation.

- **MSc Thesis: JoLT KV-Cache Compression via Tensor Optimization**

Mar 2026 - Present

- Developed JoLT, a training-free KV-cache compressor combining partial Tucker decomposition, rotated low-bit residuals, and joint Lagrangian allocation of ranks and residual bit-widths under a storage budget.

- Achieved near-lossless 2x–3x compression (perplexity drift below 0.2%) across five language models in four architecture families. Found that higher compression becomes more viable at longer contexts: on LLaMA-3.1-8B at 64K context, 4x and 5x compression reduced RULER accuracy by just 0.9 and 2.4 percentage points, respectively.
- Developed FlashJoLT, a faster variant of JoLT using randomized SVD to accelerate KV-cache compression by up to 3.97x. Separately implemented a fused Triton attention kernel operating on compressed KV-cache factors, reducing peak decode memory by 1.43–4.73x at the cost of higher inference latency.

Publications

- **A JoLT for the KV Cache** arXiv:2607.12550, R. Krishnan and V. Schulz; In review for ICLR 2027 *2026*
- **Tensor Decomposition of Transformer Key-Value Caches** arXiv:2609.28029, R. Krishnan and V. Schulz *2026*
- **Glaucoma Detection from Retinal Fundus Images** IEEE ICCSP 2020 *2020*

Education

- **Master of Science in Data Science** *Oct 2024 - Present*
  Universität Trier, Germany Grade: 1.5 (German Scale)
- **Bachelor of Technology in Computer Science and Engineering** *2016-20*
  Amrita Vishwa Vidyapeetham, India Grade: 1.37 (Top 10 in cohort)

Skills

Programming: Python, Java, SQL, R, Rust (basic), C++ (basic)
AI Safety & Evaluation: Specification Gaming, Reward Misspecification, CoT Faithfulness, Linear Probes, Activation Steering, Safety Monitoring, UK AISI Inspect
Reinforcement Learning: GRPO / RLVR, Reward Design, TRL, LoRA / PEFT, Calibration, Selective Prediction
ML Systems & Optimization: LLM Inference, KV-Cache Compression, Tensor Decomposition, Lagrangian Optimization, Triton, vLLM
ML & Research: PyTorch, Hugging Face Transformers, Scikit-learn, Bootstrap Confidence Intervals, Grouped Cross-Validation, Ablations
Infrastructure: Docker, GitHub Actions, FastAPI, GCP, Azure, Weights & Biases
Languages: English (C2), German (B1)

Experience

- **Student Research Assistant (studentische Hilfskraft)** *Apr 2026 - Present*
  Universität Trier, Dept. of Mathematics Trier, Germany
  - Support ongoing KV-cache compression and numerical optimization research under Prof. Dr. Volker Schulz.
  - Earlier phase (summer semester 2026): graded weekly Numerical Optimization exercise sheets (line-search and trust-region methods, KKT conditions, numerical linear algebra) and gave written feedback across the course cohort.

- **Software Developer (SDE II)** *Mar 2024 - Sep 2024*
  SAP Labs India Pvt. Ltd Bengaluru, India
  - Designed a rule-based declarative content-matching framework over SAP's cloud-integration layer (iFlow) that lets country teams configure XML field mappings between the SAP system and tax-authority payloads with no custom code; cut new-country onboarding from 10 days of development to about 3 days of configuration (a 70% reduction). Adopted by Spain, India, and Chile localization teams.
  - Built an iFlow bulk-retrieval algorithm on the Document Consistency Cockpit Spain reporting path, replacing about 100 paginated API calls with a single request and cutting report-generation time by about 30%.

- **Associate Software Developer (SDE I)** *July 2020 - Mar 2024*
  SAP Labs India Pvt. Ltd Bengaluru, India
  - Owned one of three core modules of the SAP Document Consistency Cockpit (tax compliance) end-to-end: designed the schema, the factory-class extension model for new document types and countries, and the retry and idempotency layer; reviewed with the lead architect and delegated implementation to two engineers.
  - Load-tested it at 10M-100M document cycles and took it to production for Spain (about 10-20k customers, up to about 10M documents/day) and India.
  - Initiated and led a Turkey error-handling overhaul on SAP Document and Reporting Compliance (DRC): categorized two years of incident data by root cause and shipped code fixes plus improved error messages and customer-facing FAQs. Incident volume fell about 88% over the year post-rollout (dashboard-measured) and the customer support rating rose from 2-3 to 4-5 on a 5-point scale.